Inequalities — problem 7.46

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Write each rational number from $(0, 1]$ as a fraction a/b with $\gcd(a, b) = 1$, and cover a/b with the interval

$$\left[\frac{a}{b} - \frac{1}{4b^2},\ \frac{a}{b} + \frac{1}{4b^2}\right].$$

Prove that the number $\sqrt{2}/2$ is not covered.